

MIDTERM I—SOLUTIONS

ECO403/503—INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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Instructions. The exam has a total of 120 points. I will grade it over 100. The points of each question is shown in boldface. Please use this information to allocate your time efficiently. Start writing your answers in the answer sheet provided but you can use as much paper as needed. The answer sheet has page number 1, so be sure to number any extra page that you use consecutively. Write your answers neatly and cross anything that you don't want me to consider. At the end of the exam I will staple all your papers.

1. **(30 points.)** *Choice under Uncertainty.* For any set X , let $|X|$ denote its *cardinality*, i.e. the number of its elements. Suppose that X is a set of prizes, and that $x \in X$ denotes a prize. A decision-maker (DM) says that, if he knew that the true state of the world was in some set T , then we would weakly prefer a lottery f over another lottery g ($f \succeq_T g$) if and only if

$$\sum_{s \in T} |\{x | f(x|s) > 0\}| \geq \sum_{s \in T} |\{x | g(x|s) > 0\}|.$$

- (a) **(6 points.)** Interpret the condition that defines the attitude of the decision maker regarding lotteries.
- (b) **(12 points.)** Do the DM's preferences satisfy the Continuity Axiom?

Axiom (Continuity). If $f \succeq_S g$ and $g \succeq_S h$, then $\exists \gamma \in [0, 1]$ and a lottery g such that $g \sim_S \gamma f + (1 - \gamma)h$.

- (c) **(12 points.)** Do the DM's preferences satisfy the Objective Substitution Axioms?

Axiom (Objective Substitution and Strict Objective Substitution). If $e \succeq_S f$ and $g \succeq_S h$ and $\alpha \in [0, 1]$, then $\alpha e + (1 - \alpha)g \succeq_S \alpha f + (1 - \alpha)h$.

If $e \succ_S f$ and $g \succeq_S h$ and $\alpha \in (0, 1]$, then $\alpha e + (1 - \alpha)g \succ_S \alpha f + (1 - \alpha)h$.

Solution

These are suggested solutions. Note that some of them go into considerable more detail than what I expect to see in your exams.

- (a) The decision maker cares about the number of prizes that have positive probability at each state $s \in T$ when comparing between two lotteries. Intuitively, if lottery f puts positive probability to a larger set of prizes than lottery g for states $s \in T$ then f is weakly preferred to g .
- (b) No. Let $\Omega = \{s\}$, i.e. the set of states consist of a single state; and let $X = \{z_1, z_2\}$. Suppose that $[z_1] \succ [z_2]$ and let f_t be the lottery such that:

$$f_t = t[z_1] + (1 - t)[z_2], \quad (1)$$

i.e. a lottery that gives z_1 with probability t and z_2 with probability $1 - t$. Then we have that for every $n \in \mathbb{N}$

$$f_{1/n} \succ f_1 \succ f_0.$$

However, we have that for any $\gamma \in (0, 1]$ and any $n \in \mathbb{N}$

$$\gamma f_{1/n} + (1 - \gamma)f_0 = \frac{\gamma}{n}[z_1] + \left(1 - \frac{\gamma}{n}\right)[z_2] \sim f_{1/n},$$

and for $\gamma = 0$, we have that $\gamma f_{1/n} + (1 - \gamma)f_0 \sim f_0$. Thus, there doesn't exist any $\gamma \in [0, 1]$ that could satisfy the conclusion of the Continuity axiom.

(c) No. Consider the same lottery f_t constructed in equation (1). Then $f_1 \sim f_0$ but for any $t \in (0, 1)$

$$f_t = tf_1 + (1-t)f_0 > f_1, \quad \text{and} \quad f_t > f_0$$

2. (70 points + extra 20 points.) *Procurement Contracts.* A government offers a firm a military contractor to buy $q \geq 0$ units of military hardware, in exchange for a payment of $t \geq 0$. The contractor's constant unit cost is private information to it and is either high (c_h) or low (c_ℓ) with $c_h > c_\ell$. The government's beliefs about the military contractor's cost is given by $\Pr(c_\ell) = p \in (0, 1)$. If the firm with cost c_i , $i = h, \ell$, accepts the contract (t_i, q_i) , its profits is $t_i - c_i q_i$, whereas if it rejects it receives zero profit. The payoff for the government is given by

$$\ln(q) - t.$$

- (a) (20 points.) Assume that the government observes the firm's type. Find the First Best Contract.
- (b) (30 points.) Now assume that the government cannot observe the firm's type. Find the optimal contract. Compare with the first best.
- (c) (20 points.) Assume that now the government faces a contractor with *three possible* types: high, medium and low ($c_h > c_m > c_\ell$). The beliefs of the government regarding these costs are: $\Pr(c_\ell) = p_\ell \in (0, 1)$, $\Pr(c_m) = p_m \in (0, 1)$ and $\Pr(c_h) = p_h \in (0, 1)$. Characterize as much as you can the optimal contract in this case.
- (d) (20 points.) *This one is harder.* Recall that monotonicity of the allocation is an important consequence of incentive compatibility. In the solution that you found in part (c), can you find conditions on the probabilities and/or the costs such that the solution is monotonic? If the conditions you propose fail, what would you do to alter the solution such that it is monotonic?

Solution

These are suggested solutions. Note that some of them go into considerable more detail than what I expect to see in your exams, in particular with respect to the simplification of the principal's problem.

- (a) The government's problems are, for each $i = \ell, h$

$$\begin{aligned} & \max_{(q_i, t_i)} \ln(q_i) - t_i \\ & \text{s.t.} \quad t_i - c_i q_i \geq 0 \\ & \quad q_i \geq 0. \end{aligned}$$

At the solution, the constraints must hold with equality, i.e.

$$t_i = c_i q_i.$$

Substituting into the objective function, the problem becomes:

$$\max_{q_i \geq 0} \ln(q_i) - c_i q_i$$

At the solution, we must have $q_i > 0$ since

$$\lim_{q_i \rightarrow 0} \ln(q_i) - c_i q_i = -\infty, \quad i = \ell, h.$$

Therefore, the following first order conditions must hold:

$$\frac{1}{q_i} = c_i, \quad i = \ell, h.$$

We can solve them as

$$q_i = \frac{1}{c_i}, \quad i = \ell, h.$$

Transfers are therefore

$$t_i = 1, \quad i = \ell, h.$$

(b) The government's problem is now

$$\begin{aligned} \max_{(q_i, t_i)_{i=\ell, h}} \quad & p(\ln(q_\ell) - t_\ell) + (1-p)(\ln(q_h) - t_h) \\ \text{s.t.} \quad & t_\ell - c_\ell q_\ell \geq 0 \\ & t_h - c_h q_h \geq 0 \\ & t_\ell - c_\ell q_\ell \geq t_h - c_\ell q_h \\ & t_h - c_h q_h \geq t_\ell - c_h q_\ell \\ & q_\ell \geq 0, q_h \geq 0 \end{aligned}$$

The following must be true at the solution:

- i. Participation constraint of the high cost holds with equality, i.e. $t_h - c_h q_h = 0$.

Proof. Suppose, towards a contradiction, that $t_h - c_h q_h > 0$. Then, the incentive compatibility constraint of the low cost implies that

$$t_\ell - c_\ell q_\ell \geq t_h - c_\ell q_h > t_h - c_h q_h > 0$$

This cannot be a solution since the government can decrease both t_ℓ and t_h by the same small amount without violating any of the constraints and increase its payoff. $\square$

- ii. Participation constraint of the low cost does not bind at the solution.

Proof. IC constraint of the low cost and the PC constraint of the high cost imply that

$$t_\ell - c_\ell q_\ell \geq t_h - c_\ell q_h > t_h - c_h q_h \geq 0.$$

$\square$

- iii. Incentive compatibility of the low cost holds with equality, i.e. $t_\ell - c_\ell q_\ell = t_h - c_\ell q_h$.

Proof. Suppose, towards a contradiction, that $t_\ell - c_\ell q_\ell > t_h - c_\ell q_h$. Then,

$$t_\ell - c_\ell q_\ell > t_h - c_\ell q_h > t_h - c_h q_h \geq 0$$

Therefore, the government can decrease t_ℓ by some small amount without violating any of the constraints and increase its payoff. $\square$

- iv. $q_\ell \geq q_h$.

Proof. The two IC constraints can be written as

$$c_\ell(q_\ell - q_h) \leq t_\ell - t_h \leq c_h(q_\ell - q_h)$$

Since $c_\ell < c_h$, we must have that $q_\ell \geq q_h$. $\square$

- v. Incentive compatibility of the high cost can be ignored.

Proof. Since the IC constraint for the low cost holds with equality and $q_\ell \geq q_h$, we have

$$t_\ell - t_h = c_\ell(q_\ell - q_h) \leq c_h(q_\ell - q_h)$$

which implies

$$t_h - c_h q_h \geq t_\ell - c_h q_\ell$$

$\square$

Therefore, the government's problem is equivalent to the following relaxed problem

$$\begin{aligned} \max_{(q_i, t_i)_{i=\ell, h}} \quad & p(\ln(q_\ell) - t_\ell) + (1-p)(\ln(q_h) - t_h) \\ \text{s.t.} \quad & t_h - c_h q_h = 0 \\ & t_\ell - c_\ell q_\ell = t_h - c_\ell q_h \\ & q_\ell \geq q_h \\ & q_\ell \geq 0, q_h \geq 0 \end{aligned}$$

From the equality constraints we have

$$\begin{aligned} t_h &= c_h q_h \\ t_\ell &= c_\ell q_\ell + (c_h - c_\ell) q_h \end{aligned}$$

Substituting into the objective we obtain

$$\begin{aligned} \max_{q_\ell, q_h} & p(\ln(q_\ell) - c_\ell q_\ell - (c_h - c_\ell) q_h) + (1 - p)(\ln(q_h) - c_h q_h) \\ \text{s.t.} \quad & q_\ell \geq q_h \\ & q_\ell \geq 0, q_h \geq 0 \end{aligned}$$

Let us first ignore the monotonicity condition. At the solution, we must have $q_\ell > 0$ and $q_h > 0$, and hence, the following first order conditions must hold:

$$\begin{aligned} p \left(\frac{1}{q_\ell} - c_\ell \right) &= 0 \\ -p(c_h - c_\ell) + (1 - p) \left(\frac{1}{q_h} - c_h \right) &= 0 \end{aligned}$$

The previous equations can be solved as

$$\begin{aligned} q_\ell &= \frac{1}{c_\ell} \\ q_h &= \frac{1}{c_h + \frac{p}{1-p}(c_h - c_\ell)} \end{aligned}$$

Since $c_\ell < c_h$, we have that $q_\ell > q_h$, i.e., the monotonicity condition is satisfied. Therefore, this is indeed the solution. Transfers can be calculated using

$$\begin{aligned} t_h &= c_h q_h \\ t_\ell &= c_\ell q_\ell + (c_h - c_\ell) q_h \end{aligned}$$

The low cost type produces the efficient (First-best) quantity and receives $(c_h - c_\ell) q_h$ as information rent, whereas the high cost produces less than the efficient amount and receives zero rent. High cost type's quantity is distorted to reduce the information rent of the low cost. The amount of distortion increases in the cost difference $c_h - c_\ell$, and in the fraction of low cost types p .

(c) The monopolist problem is now

$$\begin{aligned} \max_{(q_i, t_i)_{i=\ell, m, h}} & \sum_{i \in \{\ell, m, h\}} p_i (\ln(q_i) - t_i) \\ \text{s.t.} \quad & t_i - c_i q_i \geq 0, \quad i = \ell, m, h \\ & t_i - c_i q_i \geq t_j - c_j q_j, \quad i, j \in \{\ell, m, h\} \\ & q_i \geq 0, \quad i = \ell, m, h \end{aligned}$$

Following problem set 3 and the lecture slides, the following properties, which I just state without proof, must be true at the solution:

- i. The participation constraint of the high cost holds with equality, i.e. $t_h - c_h q_h = 0$.
- ii. All the other participation constraints do not bind at the solution.
- iii. The only incentive compatibility constraints that hold with equality are:

$$\begin{aligned} t_\ell - c_\ell q_\ell &= t_m - c_\ell q_m \\ t_m - c_m q_m &= t_h - c_m q_h \end{aligned}$$

- iv. All other incentive constraints do not bind at the solution.
- v. $q_\ell \geq q_m \geq q_h$.

Thus, the government's problem is equivalent to the following relaxed problem:

$$\begin{aligned}
& \max_{(q_i, t_i)_{i=\ell, m, h}} \sum_{i \in \{\ell, m, h\}} p_i (\ln(q_i) - t_i) \\
& \text{s.t.} \quad t_h - c_h q_h = 0 \\
& \quad t_\ell - c_\ell q_\ell = t_m - c_\ell q_m \\
& \quad t_m - c_m q_m = t_h - c_m q_h \\
& \quad q_\ell \geq q_m \geq q_h \\
& \quad q_i \geq 0, \quad i = \ell, m, h
\end{aligned}$$

Ignoring the monotonicity condition, we set up the Lagrangian:

$$\begin{aligned}
\mathcal{L} = & \sum_{i \in \{\ell, m, h\}} p_i (\ln(q_i) - t_i) + \lambda_h (t_h - c_h q_h) + \lambda_m (t_m - t_h - c_m q_m + c_m q_h) \\
& + \lambda_\ell (t_\ell - t_m - c_\ell q_\ell + c_\ell q_m) + \sum_{i \in \{\ell, m, h\}} \mu_i q_i
\end{aligned}$$

The KT conditions for optimality are:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial q_i} &= 0 \quad i = \ell, m, h \\
\frac{\partial \mathcal{L}}{\partial t_i} &= 0 \quad i = \ell, m, h \\
\frac{\partial \mathcal{L}}{\partial \lambda_i} &= 0 \quad i = \ell, m, h \\
\mu_i q_i &= 0, \mu_i \geq 0, q_i \geq 0 \quad i = \ell, m, h
\end{aligned}$$

For the moment, we will focus on solutions such that $q_i > 0$, which implies that $\mu_i = 0$ for all i . For $i = h$ we have

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial q_h} &= \frac{p_h}{q_h} - \lambda_h c_h + \lambda_m c_m = 0 \\
\frac{\partial \mathcal{L}}{\partial t_h} &= -p_h + \lambda_h - \lambda_m = 0 \\
\frac{\partial \mathcal{L}}{\partial \lambda_h} &= t_h - c_h q_h = 0
\end{aligned}$$

For $i = m$ we have

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial q_m} &= \frac{p_m}{q_m} - \lambda_m c_m + \lambda_\ell c_\ell = 0 \\
\frac{\partial \mathcal{L}}{\partial t_m} &= -p_m + \lambda_m - \lambda_\ell = 0 \\
\frac{\partial \mathcal{L}}{\partial \lambda_m} &= t_m - t_h - c_m q_m + c_m q_h = 0
\end{aligned}$$

For $i = \ell$ we have

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial q_\ell} &= \frac{p_\ell}{q_\ell} - \lambda_\ell c_\ell = 0 \\
\frac{\partial \mathcal{L}}{\partial t_\ell} &= -p_\ell + \lambda_\ell = 0 \\
\frac{\partial \mathcal{L}}{\partial \lambda_\ell} &= t_\ell - t_m - c_\ell q_\ell + c_\ell q_m = 0
\end{aligned}$$

Thus we have that $\lambda_\ell = p_\ell$, $\lambda_m = p_m + p_\ell$ and that $\lambda_h = p_h + p_m + p_\ell$. From this we get that:

$$q_\ell = \frac{1}{c_\ell}$$

$$q_m = \frac{1}{c_m + \frac{p_\ell}{p_m}(c_m - c_\ell)}$$

$$q_h = \frac{1}{c_h + \frac{p_m + p_\ell}{p_h}(c_h - c_m)}$$

Notice that because $1/x > 0$ for all $x > 0$ and all costs and probabilities are positive in the problem, we will not have corner solutions. Thus we were right to assume initially that the solutions would be strictly positive.

Once again we have that there is no distortion at the top, i.e., q_ℓ is equal to the efficient quantity. For types with medium and high costs, there will be a distortion in quantity.

Finally, transfers can be obtained from the following equalities:

$$t_h = c_h q_h$$

$$t_m = c_m(q_m - q_h) + t_h$$

$$t_\ell = c_\ell(q_\ell - q_m) + t_m$$

- (d) Note that in the solution above $q_\ell > q_m$. However, it not obvious when we will have $q_m \geq q_h$. We can proceed in two ways:

i. *Necessary and sufficient condition:*

$$q_m \geq q_h \Leftrightarrow \frac{1}{c_m + \frac{p_\ell}{p_m}(c_m - c_\ell)} \geq \frac{1}{c_h + \frac{p_m + p_\ell}{p_h}(c_h - c_m)}$$

$$\Leftrightarrow c_m + \frac{p_\ell}{p_m}(c_m - c_\ell) \leq c_h + \frac{p_m + p_\ell}{p_h}(c_h - c_m)$$

$$\Leftrightarrow \frac{p_\ell}{p_m}(c_m - c_\ell) \leq (c_h - c_m) \left(\frac{p_m + p_\ell}{p_h} \right)$$

$$\Leftrightarrow \frac{\frac{p_\ell}{p_m}}{\frac{p_h + p_m + p_\ell}{p_h}} = \frac{p_h p_\ell}{p_m} \leq \frac{c_h - c_m}{c_m - c_\ell}$$

ii. *Sufficient conditions:* The following inequalities must hold

$$\frac{p_\ell}{p_m} \leq \frac{p_m + p_\ell}{p_h} \quad c_m - c_\ell \leq c_h - c_m$$

The sufficient conditions are the ones usually found in the literature: for this problem, where the ordering of costs is $c_h > c_m > c_\ell$, we need an increasing inverse hazard rate and that the difference in costs be increasing. (Check the last slide of the last batch of lecture notes and my solutions for PS3.)

Now, for the hard part of this question, I will show you how to adjust the previous solution in case the previous conditions I identified fail. We know that we will always have $q_\ell > q_m$. But if the previous conditions fail, then we will have that the monotonicity constraint between the medium cost type and the high cost type is binding, i.e. $q_m = q_h = \tilde{q}$. In this case, the relaxed problem is adjusted as follows:

$$\max_{(q_\ell, t_\ell), (\tilde{q}, \tilde{t})} (p_h + p_m)(\ln(\tilde{q}) - \tilde{t}) + p_\ell(\ln(q_\ell) - t_\ell)$$

$$\text{s.t. } \tilde{t} - c_h \tilde{q} = 0$$

$$t_\ell - c_\ell q_\ell = \tilde{t} - c_m \tilde{q}$$

$$q_\ell \geq 0, \tilde{q} \geq 0.$$

Substituting the equality constraints into the objective yields:

$$\max_{q_\ell \geq 0, \tilde{q} \geq 0} (p_h + p_m)(\ln(\tilde{q}) - c_h \tilde{q}) + p_\ell(\ln(q_\ell) - \tilde{q}(c_h - c_\ell) - c_\ell q_\ell)$$

The first order conditions yield:

$$\begin{aligned} (p_h + p_m) \left(\frac{1}{\tilde{q}} - c_h \right) - p_\ell(c_h - c_\ell) &= 0 \\ p_\ell \left(\frac{1}{q_\ell} - c_\ell \right) &= 0 \end{aligned}$$

The previous two equations can be solved for the quantities as

$$\begin{aligned} \tilde{q} &= \frac{1}{c_h + \frac{p_\ell}{p_h + p_m}(c_h - c_\ell)} \\ q_\ell &= \frac{1}{c_\ell} \end{aligned}$$

Finally, transfers can be obtained from the equality restrictions

$$\begin{aligned} \tilde{t} &= c_h \tilde{q} \\ t_\ell &= \tilde{q}(c_h - c_\ell) + c_\ell q_\ell. \end{aligned}$$